Name: _____ Total: 18 points

As usual, let $p_{\text{data}}$ be the unknown target distribution, and $x^{(i)} \in \mathbb{R}^d$ be iid samples according to $p_{\text{data}}$ that are given to us. Let $X_1 \sim p_{\text{data}}$ and $X_0 \sim \mu$, where $\mu$ is an easy-to-sample distribution, e.g., a standard normal distribution in $\mathbb{R}^d$. In flow matching, the goal is to learn a vector field, $v_t(x)$, such that its flow $\varphi^t(x)$ satisfies

$$\varphi^1_\sharp \mu = p_{\text{data}}.$$

To do this, we consider the intermediate marginal densities, $\rho_t$, which satisfy the continuity equation, as a total expectation:

$$\rho_t(x) = \mathbb{E}_{z \sim q_Z} p_{X_t|Z}(x|z), \tag{1}$$

where $Z = (X_0, X_1)$ is chosen according to a joint distribution $q_Z$. Here, $p_{X_t|Z}$ is the conditional density of $X_t$.

1. Let $p_{X_t|Z} = \mathcal{N}(m_t, \sigma_t^2 \operatorname{Id})$, with $m_t((x_0, x_1)) = tx_1 + (1-t)x_0$. Then, the path connecting $X_0$ and $X_1$ where $(X_0, X_1) \sim q_Z$, becomes a straight line for any choice of coupling $q_Z$. True or False? Justify your answer. (2 points)

2. True or false: the OT-CFM flow $\varphi^1$ is the same as the Brenier transport map (2 points)

3. Match the learned vector fields from three different conditional flow matching algorithms: CFM, OT-CFM and stochastic interpolants. In stochastic interpolants, $q_Z = \rho_0(x_0)\,\rho_1(x_1)$ and $m_t((x_0, x_1)) = \cos(\pi t/2)x_0 + \sin(\pi t/2)x_1$. The distribution $p_{\text{data}}$ is the following 2 moons distribution:

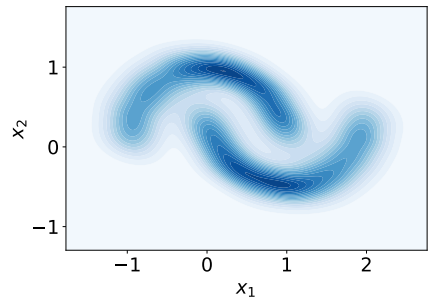


Figure 1: $p_{\text{data}}$

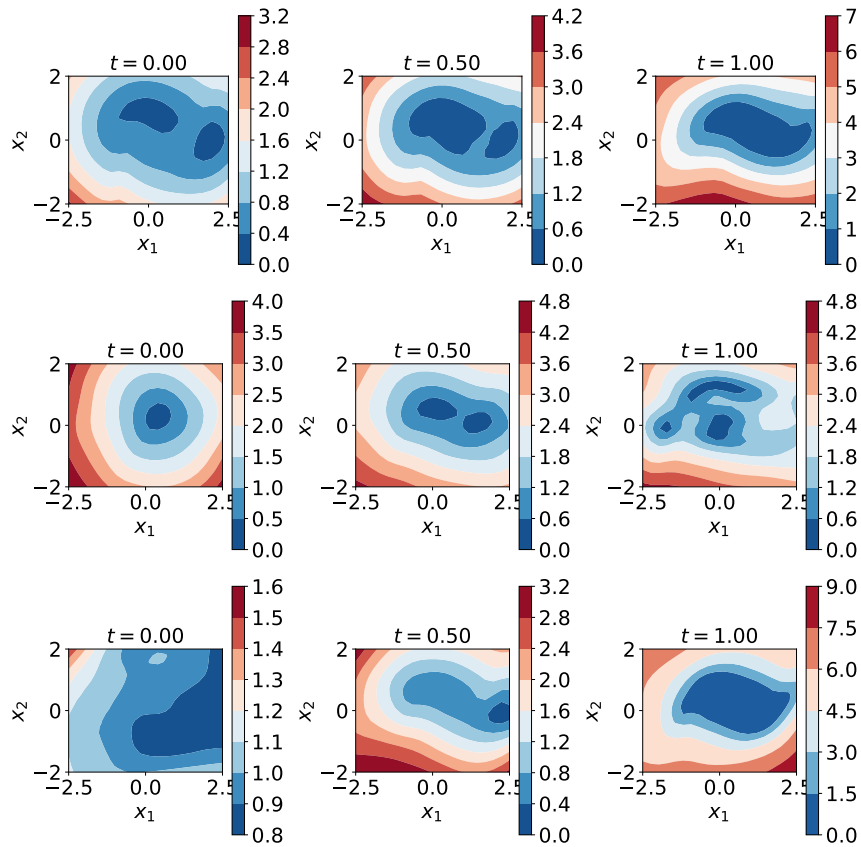


Figure 2: Row 1: ....., Row 2: ....., Row 3:.....

3 points for the correct match, 3 for the correct reasoning.

4. Match the learned paths below. The starting point, $x_0$, is marked as a square and the end point, $x_1$, is marked as a circle. Justify your answer (5 points)

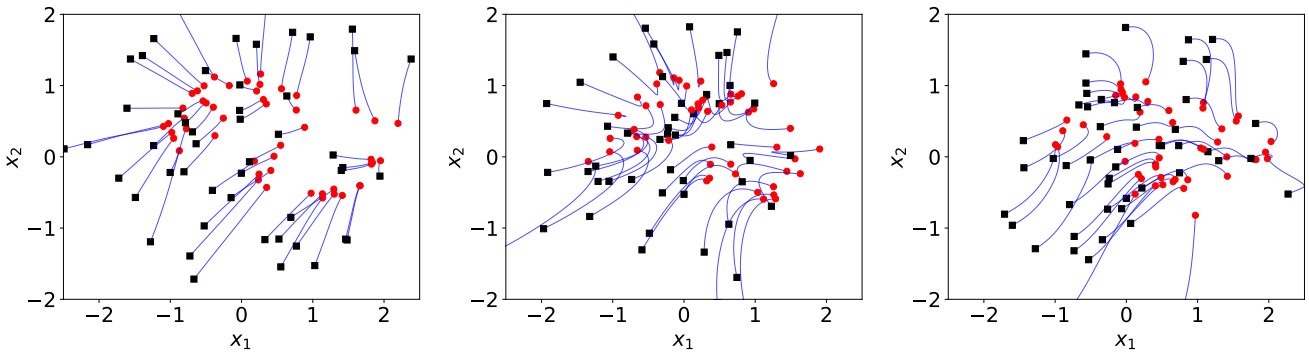


Figure 3: Left:....., Center:....., Right:.....

5. Now consider a new process: $X_0 = \alpha_t X_1 + \sigma_t \epsilon$, where $\epsilon \sim \mathcal{N}(0, \mathrm{Id})$. This is a noising process in a diffusion model. What is the conditional expectation of $p_{\text{data}}$ given a noised sample, $X_0$? (1 point) If $\alpha_t$ decreases with $t$ and $\sigma_t$ increases with $t$, when is the conditional expectation a good approximation of a sample $X_1 \sim p_{\text{data}}$, for small or large $t$? Explain (2 points)